Robust Principal Components of Curve Changes: S-estimation of the Principal Subspace and its Implementation by Weighted EMPCA

Technical note

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Abstract

We study principal component analysis (PCA) of daily changes of a term structure when a single day is an order of magnitude larger than typical days. We first state classical PCA in its original variance-maximisation form (Hotelling, 1933) and prove its equivalence with the best affine rank-q approximation (Pearson, 1901; Eckart and Young, 1936). This equivalence shows that classical PCA minimises a non-robust scale (a root mean square) of the residual norms, and the influence function of the eigenvectors (Critchley, 1985) quantifies the damage done by one large day. We then present the robust alternative of Maronna (2005), implemented as `pcaRobS` in the `R` package `RobStatTM` that accompanies Maronna et al. (2019): the principal subspace is the minimiser of a bounded M-scale of residual norms. We derive its estimating equations, the iteratively reweighted algorithm and a monotone-descent theorem. We then show that the same estimator is obtained by wrapping the weighted EMPCA of Bailey (2012) in an outer reweighting loop, with Bailey's weights set to $W_t\,\omega_{tj}$, and we state exactly when that combined algorithm is guaranteed to decrease the objective. Two results bear directly on practice: (i) a residual-based estimator protects the *subspace* but cannot protect the individual components against an outlier that lies *inside* the subspace (a large pure level/slope day), which requires a second, robust rotation step; (ii) for norm-based down-weighting, the weight entering the scatter matrix must decay like $\|\tilde{x}_t\|^{-2}$, not $\|\tilde{x}_t\|^{-1}$. All claims are checked numerically, including an exact replication of `pcaRobS`.

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1 Setting and notation

We observe n daily changes of a curve on p fixed tenors, $x_t \in \mathbb{R}^p$, $t = 1, \dots, n$, collected as rows of $X \in \mathbb{R}^{n\times p}$. Throughout, $q \in \{1, \dots, p-1\}$ is the number of components retained. $\|\cdot\|$ is the Euclidean norm on vectors and the spectral norm on matrices; $\|\cdot\|_F$ is the Frobenius norm. The Stiefel manifold is

$$\mathcal{V}_{p,q} = \{B \in \mathbb{R}^{p\times q} : \ B^\top B = I_q\}.$$

For $B \in \mathcal{V}_{p,q}$ we write $P_B = BB^\top$ for the orthogonal projector onto $\operatorname{span}(B)$ and $P_\perp = I_p - P_B$ (the dependence on B is suppressed when clear). For a centre $\mu \in \mathbb{R}^p$ we write $\tilde{x}_t = x_t - \mu$ and define the *residual norm* of day t:

$$r_t(\mu, B)^2 = \|P_\perp(x_t - \mu)\|^2 = \|\tilde{x}_t\|^2 - \|B^\top \tilde{x}_t\|^2. \tag{1}$$

Eigenvalues of a symmetric matrix are ordered $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$, with orthonormal eigenvectors $v_1, \dots, v_p$; $V_q = [v_1, \dots, v_q]$.

The motivating situation is one day t_o with $\|\tilde{x}_{t_o}\| \approx 18 \times \operatorname{median}_t \|\tilde{x}_t\|$. We call a day *in-span* if its deviation lies (almost) in the true principal subspace (e.g. a very large parallel shift or steepening), and *off-span* otherwise.

2 Classical PCA

2.1 Population definition (Hotelling)

Let x be a random vector in $\mathbb{R}^p$ with mean $\mu = \mathbb{E}x$ and covariance $\Sigma = \mathbb{E}[(x-\mu)(x-\mu)^\top]$, with spectral decomposition $\Sigma = \sum_{j=1}^p \lambda_j v_j v_j^\top$.

Definition 2.1 (Principal components; Hotelling, 1933)**.** The first principal direction is any

$$a_1 \in \arg\max_{\|a\|=1} \operatorname{Var}(a^\top x) = \arg\max_{\|a\|=1} a^\top \Sigma a.$$

For $k \geq 2$, the k-th principal direction is any

$$a_k \in \arg\max\{a^\top \Sigma a : \ \|a\| = 1, \ \operatorname{Cov}(a^\top x, a_l^\top x) = 0, \ l < k\}.$$

The k-th principal component is the random variable $z_k = a_k^\top (x - \mu)$ and its variance is $\operatorname{Var}(z_k)$.

Theorem 2.2 (Hotelling)**.** *Assume $\lambda_1 > \lambda_2 > \cdots > \lambda_q > \lambda_{q+1} \geq 0$ and $\lambda_q > 0$. Then for $k \leq q$, $a_k = \pm v_k$ and $\operatorname{Var}(z_k) = \lambda_k$. The components $z_1, \dots, z_q$ are uncorrelated and $\sum_{k\leq q} \lambda_k / \operatorname{tr}\Sigma$ is the proportion of total variance they explain.*

Proof. Write $a = \sum_j \alpha_j v_j$ with $\sum_j \alpha_j^2 = 1$. Then $a^\top \Sigma a = \sum_j \lambda_j \alpha_j^2 \le \lambda_1$, with equality if and only if $\alpha_j = 0$ whenever $\lambda_j < \lambda_1$; since $\lambda_1 > \lambda_2$, $a_1 = \pm v_1$. Suppose $a_l = \pm v_l$ for $l < k$. Then $\mathrm{Cov}(a^\top x, a_l^\top x) = a^\top \Sigma v_l = \lambda_l \alpha_l$, so (as $\lambda_l > 0$) the constraint is $\alpha_l = 0$ for $l < k$. Then $a^\top \Sigma a = \sum_{j \ge k} \lambda_j \alpha_j^2 \le \lambda_k$, with equality iff $a = \pm v_k$ (using $\lambda_k > \lambda_{k+1}$). Finally $\mathrm{Var}(z_k) = v_k^\top \Sigma v_k = \lambda_k$ and $\mathrm{Cov}(z_k, z_l) = v_k^\top \Sigma v_l = 0$. □

2.2 Sample version

Replacing (μ, Σ) by the sample mean and covariance,

$$\bar{x} = \frac{1}{n} \sum_{t=1}^{n} x_t, \qquad S = \frac{1}{n-1} \sum_{t=1}^{n} (x_t - \bar{x})(x_t - \bar{x})^\top, \tag{2}$$

gives the sample principal directions $\hat{v}_k$ (eigenvectors of S), sample variances $\hat{\lambda}_k$, scores $\hat{z}_{tk} = \hat{v}_k^\top (x_t - \bar{x})$, and the explained-variance ratio $\sum_{k \le q} \hat{\lambda}_k / \operatorname{tr} S$. Numerically one uses the SVD of the centred data matrix $\tilde{X} = X - \mathbf{1}\bar{x}^\top = UDG^\top$: then $\hat{v}_k$ is the k-th column of G and $\hat{\lambda}_k = d_k^2/(n-1)$.

2.3 Equivalent characterisations

The robust estimators below are built on a characterisation of PCA that is equivalent to Definition 2.1 but is not its definition. We prove the equivalence.

Lemma 2.3 (Ky Fan maximum principle; Fan, 1949)**.** *Let $S \in \mathbb{R}^{p \times p}$ be symmetric with eigenpairs (λ_j, v_j). Then*

$$\max_{B \in \mathcal{V}_{p,q}} \operatorname{tr}(B^\top S B) = \sum_{j=1}^{q} \lambda_j,$$

attained at $B = V_q$. If $\lambda_q > \lambda_{q+1}$, the maximisers are exactly the B with $\operatorname{span}(B) = \operatorname{span}(V_q)$.

Proof. Let $h_j = \|B^\top v_j\|^2$. Then $\operatorname{tr}(B^\top S B) = \sum_j \lambda_j v_j^\top B B^\top v_j = \sum_j \lambda_j h_j$. Since $BB^\top$ is an orthogonal projector, $0 \le h_j \le \|v_j\|^2 = 1$, and $\sum_j h_j = \|B^\top [v_1, \dots, v_p]\|_F^2 = \|B\|_F^2 = q$. Maximising the linear function $\sum_j \lambda_j h_j$ over $\{0 \le h_j \le 1,\ \sum_j h_j = q\}$ with $\lambda_1 \ge \cdots \ge \lambda_p$ puts $h_j = 1$ for $j \le q$ and 0 otherwise, giving $\sum_{j \le q} \lambda_j$. If $\lambda_q > \lambda_{q+1}$, any other feasible h gives a strictly smaller value, so $h_j = 1$ $(j \le q)$ is necessary, i.e. $v_1, \dots, v_q \in \operatorname{span}(B)$; since $\dim \operatorname{span}(B) = q$, $\operatorname{span}(B) = \operatorname{span}(V_q)$. □

Theorem 2.4 (Equivalent characterisations of classical PCA)**.** *Assume $\lambda_q(S) > \lambda_{q+1}(S)$. For $(\mu, B) \in \mathbb{R}^p \times \mathcal{V}_{p,q}$ let*

$$L(\mu, B) = \sum_{t=1}^{n} r_t(\mu, B)^2 = \sum_{t=1}^{n} \|P_\perp (x_t - \mu)\|^2.$$

The following are equivalent:

(i) $\operatorname{span}(B) = \operatorname{span}(\hat{v}_1, \dots, \hat{v}_q)$ *(Hotelling, sample version);*

(ii) *B maximises $\operatorname{tr}(B^\top S B)$ over $\mathcal{V}_{p,q}$;*

(iii) *for some μ, (μ, B) minimises L (best affine q-dimensional fit; Pearson, 1901);*

(iv) *$\tilde{X} B B^\top$ is a best rank-q approximation of $\tilde{X}$ in Frobenius norm (Eckart and Young, 1936).*

Moreover, for fixed B the minimisers of $L(\cdot, B)$ are exactly $\mu \in \bar{x} + \operatorname{span}(B)$.

Proof. Centre. For fixed B, $L(\mu, B) = \sum_t \|P_\perp x_t - P_\perp \mu\|^2$ is a least-squares problem in $P_\perp \mu$, minimised iff $P_\perp \mu = P_\perp \bar{x}$, i.e. $\mu \in \bar{x} + \mathrm{span}(B)$. *(iii)⇔(ii).* At such μ, using (1), $\min_\mu L(\mu, B) = \sum_t \|x_t - \bar{x}\|^2 - \sum_t \|B^\top (x_t - \bar{x})\|^2 = (n-1)\{\mathrm{tr}\, S - \mathrm{tr}(B^\top S B)\}$. *(ii)⇔(i)* is Lemma 2.3. *(iii)⇔(iv).* A matrix M of rank $\le q$ has its rows in some q-dimensional subspace $\mathrm{span}(B)$; for fixed $\mathrm{span}(B)$, $\|\tilde{X} - M\|_F^2$ is minimised row by row by the orthogonal projection, $M = \tilde{X} B B^\top$, with value $L(\bar{x}, B)$. Hence minimising over rank-q matrices is minimising $L(\bar{x}, \cdot)$ over $\mathcal{V}_{p,q}$. □

Corollary 2.5 (Weighted version)**.** *Let $W_1, \dots, W_n \ge 0$ with $\sum_t W_t > 0$, and set $\bar{x}_W = \sum_t W_t x_t / \sum_t W_t$, $C_W = \sum_t W_t (x_t - \bar{x}_W)(x_t - \bar{x}_W)^\top$ and $L_W(\mu, B) = \sum_t W_t r_t(\mu, B)^2$. If $\lambda_q(C_W) > \lambda_{q+1}(C_W)$, the minimisers of L_W are exactly the pairs with $\mathrm{span}(B)$ equal to the top-q eigenspace of C_W and $\mu \in \bar{x}_W + \mathrm{span}(B)$; the minimum value is $\mathrm{tr}\, C_W - \sum_{j \le q} \lambda_j(C_W)$.*

Proof. Repeat the proof of Theorem 2.4 with weighted sums (equivalently, apply it to the rows $\sqrt{W_t}(x_t - \bar{x}_W)$). □

Remark 2.6 (Why (iii) is the right starting point). By (iii), classical PCA chooses (μ, B) to minimise

$$s_{\mathrm{LS}}(r_1, \dots, r_n) = \left(\tfrac{1}{n} \textstyle\sum_t r_t^2 \right)^{1/2},$$

the root mean square of the residual norms. s_{LS} is a measure of scale that a single observation can make arbitrarily large. Robust PCA in the sense of Maronna (2005) keeps (iii) and replaces s_{LS} by a robust scale (Section 5). Note that (iii) estimates a *subspace*; the individual directions inside it come from an additional step (the eigenvectors of S restricted to the subspace). This distinction matters in Section 6.

3 How one large day distorts classical PCA

3.1 Influence function

Let F be the distribution of x, and $F_\varepsilon = (1 - \varepsilon)F + \varepsilon \delta_y$ a contamination by a point mass at y. The influence function of a functional T is $\mathrm{IF}(y; T, F) = \frac{d}{d\varepsilon} T(F_\varepsilon)\big|_{\varepsilon = 0}$ (Hampel et al., 1986).

Lemma 3.1 (First-order eigen-perturbation)**.** *Let Σ be symmetric with a simple eigenvalue λ_j and unit eigenvector v_j, and $\Sigma(\varepsilon) = \Sigma + \varepsilon E + o(\varepsilon)$ with E symmetric. Then there are differentiable $\lambda_j(\varepsilon)$, $v_j(\varepsilon)$ (unit, sign fixed by $v_j^\top v_j(\varepsilon) > 0$) with*

$$\lambda_j'(0) = v_j^\top E v_j, \qquad v_j'(0) = \sum_{k \ne j} \frac{v_k^\top E v_j}{\lambda_j - \lambda_k} v_k.$$

Proof. Differentiability of a simple eigenpair is standard (Golub and Van Loan, 2013, Ch. 8). Differentiate $\Sigma(\varepsilon) v_j(\varepsilon) = \lambda_j(\varepsilon) v_j(\varepsilon)$ at 0: $E v_j + \Sigma v_j' = \lambda_j' v_j + \lambda_j v_j'$. Pre-multiplying by $v_j^\top$ gives $\lambda_j' = v_j^\top E v_j$; by $v_k^\top$, $k \ne j$, gives $v_k^\top E v_j = (\lambda_j - \lambda_k) v_k^\top v_j'$. Differentiating $\|v_j(\varepsilon)\|^2 = 1$ gives $v_j^\top v_j' = 0$. □

Proposition 3.2 (Influence of one observation on classical PCA; Critchley, 1985)**.** *Let $z = V^\top (y - \mu)$ be the population scores of the contaminating point y, and let λ_j be simple. Then*

$$\mathrm{IF}(y; \lambda_j, F) = z_j^2 - \lambda_j, \qquad \mathrm{IF}(y; v_j, F) = \sum_{k \ne j} \frac{z_j z_k}{\lambda_j - \lambda_k} v_k.$$

Both are unbounded in y and grow quadratically in $\|y - \mu\|$.

Proof. With $\mu_\varepsilon = \mu + \varepsilon(y - \mu)$, $\Sigma(F_\varepsilon) = (1-\varepsilon)\{\Sigma + (\mu - \mu_\varepsilon)(\mu - \mu_\varepsilon)^\top\} + \varepsilon(1-\varepsilon)^2 (y-\mu)(y-\mu)^\top = \Sigma + \varepsilon\{(y - \mu)(y - \mu)^\top - \Sigma\} + O(\varepsilon^2)$. Apply Lemma 3.1 with $E = (y - \mu)(y - \mu)^\top - \Sigma$, for which $v_k^\top E v_j = z_k z_j - \lambda_j \mathbf{1}\{k = j\}$. □

Two consequences matter here. First, the eigenvector v_j is pulled towards every other v_k in proportion to $z_j z_k/(\lambda_j - \lambda_k)$, so a large day that loads on two components rotates them into each other. The effect is largest for the lower components, whose eigenvalue gaps are small. Second, in a finite sample the day's share of total variance is $\pi_{t_o} = \|x_{t_o} - \bar{x}\|^2 / \sum_t \|x_t - \bar{x}\|^2$. In the simulation of Section 9 ($n = 750$, one day at 18× the median norm), $\pi_{t_o} = 0.21$: one day out of 750 accounts for about a fifth of $\operatorname{tr} S$.

3.2 In-span versus off-span outliers

Lemma 3.3 (An in-span outlier leaves the subspace invariant but rotates the components)**.** *Let S_0 be symmetric positive semidefinite with $\lambda_q(S_0) > \lambda_{q+1}(S_0)$ and top-q eigenspace U. Let $y \in U$, $\alpha > 0$ and $S_1 = S_0 + \alpha y y^\top$. Then*

(a) U is still the top-q eigenspace of S_1;

(b) for $j \le q$, $v_j(S_0)$ is an eigenvector of S_1 iff $y \perp v_j$ or $y \in \operatorname{span}(v_j)$.

Proof. (a) For $u \in U$, $S_1 u = S_0 u + \alpha y (y^\top u) \in U$, so U and (by symmetry) $U^\perp$ are invariant. On $U^\perp$, $S_1 = S_0$ (as $y \perp U^\perp$), with eigenvalues $\lambda_{q+1}, \dots, \lambda_p$. On U, $S_1 \succeq S_0$, so by Weyl's inequality its eigenvalues are $\ge \lambda_q(S_0) > \lambda_{q+1}(S_0)$. (b) $S_1 v_j = \lambda_j v_j + \alpha (y^\top v_j) y$, which is parallel to v_j iff $y^\top v_j = 0$ or $y \parallel v_j$. □

So a huge parallel-plus-steepening day does not tilt the level–slope–curvature *subspace*, but it mixes the level and slope *loadings*. As $\alpha \to \infty$, PC1 tends to $y/\|y\|$. An off-span day ($P_\perp y \neq 0$) does both: it tilts the subspace and rotates the components.

4 Robust scales

Definition 4.1 (ρ-function)**.** A ρ-function is $\rho : \mathbb{R} \to [0, 1]$ that is even, continuous, nondecreasing in $|u|$, with $\rho(0) = 0$, $\sup_u \rho(u) = 1$, and strictly increasing on $\{u \ge 0 : \rho(u) < 1\}$.

Example 4.2 (Bisquare). $\rho_c(u) = 1 - \big(1 - (u/c)^2\big)^3$ for $|u| \le c$ and $\rho_c(u) = 1$ otherwise. `pcaRobS` uses $c = 1$ (we checked that `RobStatTM::rho(u,"bisquare",cc=1)` equals this formula).

Definition 4.3 (M-scale)**.** Given $\delta \in (0, 1)$ and $r = (r_1, \dots, r_n) \in [0, \infty)^n$, the M-scale $\hat{\sigma}(r)$ is the solution $\sigma > 0$ of

$$h_r(\sigma) := \frac{1}{n} \sum_{t=1}^{n} \rho\Big(\frac{r_t}{\sigma}\Big) = \delta. \tag{3}$$

Lemma 4.4 (Existence and uniqueness)**.** *If $\#\{t : r_t > 0\} > n\delta$, (3) has a unique solution $\hat{\sigma}(r) > 0$. Moreover $\hat{\sigma}(\beta r) = \beta \hat{\sigma}(r)$ for $\beta > 0$, and $h_r(s) \le \delta \Rightarrow \hat{\sigma}(r) \le s$.*

Proof. h_r is continuous and nonincreasing on $(0, \infty)$, with $h_r(\sigma) \to \#\{r_t > 0\}/n > \delta$ as $\sigma \downarrow 0$ and $h_r(\sigma) \to 0$ as $\sigma \to \infty$; by the intermediate value theorem a solution exists. If $h_r(\sigma) \in (0, 1)$, some r_t/σ lies in the region where ρ is strictly increasing, so h_r is strictly decreasing at σ; hence the solution is unique. Equivariance follows from $h_{\beta r}(\beta\sigma) = h_r(\sigma)$. For the last claim, if $\hat{\sigma} > s$ then $h_r(s) \ge h_r(\hat{\sigma}) = \delta$; combined with $h_r(s) \le \delta$ this gives $h_r(s) = \delta$, so $s = \hat{\sigma}$ by uniqueness, a contradiction. □

Remark 4.5 (Robustness and tuning). Because $\rho \le 1$, a single observation changes the left side of (3) by at most $1/n$, whatever its size. The finite-sample breakdown point of $\hat{\sigma}$ is $\min(\delta, 1 - \delta)$ up to $O(1/n)$ (Huber, 1981; Maronna et al., 2019); $\delta = 0.5$ maximises it. By equivariance, replacing $\rho(u)$ by $\rho(u/c)$ multiplies $\hat{\sigma}$ by c *for every* r, so the minimiser in Definition 5.1 below depends on δ only. The constant c only calibrates the value of σ.

Weights. Put $g(\tau) = \rho(\sqrt{\tau})$ for $\tau \geq 0$, so that $\rho(u) = g(u^2)$ and $\rho'(u)/u = 2g'(u^2)$.

Lemma 4.6. *For the bisquare with $c = 1$, $g(\tau) = 1 - (1-\tau)^3$ on $[0,1]$ and $g = 1$ on $[1,\infty)$. g is C^1, nondecreasing and concave on $[0,\infty)$, with $g'(\tau) = 3(1-\tau)_+^2$.*

Proof. $g'(\tau) = 3(1-\tau)^2$ on $[0,1]$, which equals 0 at $\tau = 1$ and matches $g' = 0$ on $(1,\infty)$. g' is continuous, nonnegative and nonincreasing, so g is nondecreasing and concave. □

We use the normalised weight function

$$w(\tau) = \frac{g'(\tau)}{g'(0)} = (1-\tau)_+^2 \in [0,1], \tag{4}$$

which is `Wf` in `pcaRobS`. The weight of a residual r at scale σ is $w(r^2/\sigma^2)$, and it is *redescending*: it is exactly 0 when $r \geq \sigma$.

5 S-estimation of the principal subspace

5.1 Definition

Definition 5.1 (S-PCA; Maronna, 2005)**.** For $\theta = (\mu, B) \in \Theta = \mathbb{R}^p \times \mathcal{V}_{p,q}$ let $r_t(\theta)$ be as in (1) and $\hat{\sigma}(\theta) = \hat{\sigma}(r_1(\theta), \ldots, r_n(\theta))$ the M-scale of Definition 4.3. The S-estimator of the principal subspace is

$$\hat{\theta} \in \arg\min_{\theta \in \Theta} \hat{\sigma}(\theta).$$

Proposition 5.2 (Basic properties)**.** *(a) $\hat{\sigma}(\mu, B)$ depends on B only through $\operatorname{span}(B)$ and on μ only through $\mu + \operatorname{span}(B)$.*

(b) The estimator is equivariant under $x_t \mapsto \beta Q x_t + b$ with Q orthogonal, $\beta > 0$ and $b \in \mathbb{R}^p$.

(c) Replacing $\hat{\sigma}$ by s_{LS} of Remark 2.6 (formally $\rho(u) = u^2$, $\delta = 1$) gives classical PCA by Theorem 2.4.

Proof. (a) r_t depends on B through $P_\perp$, and $P_\perp(\mu + Bb) = P_\perp \mu$. (b) Under the map, $r_t(\beta Q\mu + b, QB) = \beta r_t(\mu, B)$, and $\hat{\sigma}$ is scale equivariant (Lemma 4.4). (c) is immediate. □

Property (b) is orthogonal, not affine, equivariance. This suits curve data because all tenors are in the same units.

5.2 Estimating equations

Theorem 5.3 (First-order conditions)**.** *Let ρ be as in Example 4.2 and θ^* be such that some $r_t(\theta^*) \in (0, \hat{\sigma}(\theta^*))$. Write $\sigma^* = \hat{\sigma}(\theta^*)$, $W_t^* = w(r_t(\theta^*)^2/\sigma^{*2})$ and $P^* = B^* B^{*\top}$. Then $\hat{\sigma}$ is differentiable at θ^*, and θ^* is a stationary point of $\hat{\sigma}$ on Θ iff*

$$\textstyle\sum_t W_t^* \, P_\perp^*(x_t - \mu^*) = 0, \tag{5}$$

$$P_\perp^* \, C^* \, P^* = 0, \qquad C^* = \textstyle\sum_t W_t^* (x_t - \mu^*)(x_t - \mu^*)^\top. \tag{6}$$

That is, $P_\perp^* \mu^* = P_\perp^* \bar{x}_{W^*}$ and $\operatorname{span}(B^*)$ is an invariant subspace of the weighted scatter C^*. These are exactly the stationarity conditions of the weighted least-squares criterion L_{W^*} of Corollary 2.5 with the weights frozen at W^*.

Algorithm 1 S-PCA by iteratively reweighted least squares (`RobStatTM::pcaRobS`)

Require: X, q, δ (default 0.5), tolerance ϵ (default 10^{-4}), maximum iterations K (default 100)

1: *Spherical start* (Locantore et al., 1999): μ^0 = spatial median; $u_t = (x_t - \mu^0)/\|x_t - \mu^0\|$; B^0 = top-q eigenvectors of $\sum_t u_t u_t^\top$
2: $r_t \leftarrow r_t(\mu^0, B^0)$, $\sigma \leftarrow \hat{\sigma}(r)$
3: **for** $k = 1, \ldots, K$ **do**
4: $\quad W_t \leftarrow w(r_t^2/\sigma^2)$ ▷ weights frozen for this step
5: $\quad \mu \leftarrow \sum_t W_t x_t / \sum_t W_t$
6: $\quad B_{\text{new}} \leftarrow$ top-q eigenvectors of $C_W = \sum_t W_t (x_t - \mu)(x_t - \mu)^\top$
7: $\quad r_t \leftarrow r_t(\mu, B_{\text{new}})$, $\sigma_{\text{new}} \leftarrow \hat{\sigma}(r)$
8: $\quad \Delta \leftarrow \max\{1 - \sigma_{\text{new}}^2/\sigma^2,\ \text{mean}\,|I_q - |B_{\text{new}}^\top B||\}$
9: $\quad B \leftarrow B_{\text{new}}$, $\sigma \leftarrow \sigma_{\text{new}}$; **if** $\Delta < \epsilon$ **break**
10: **end for**
11: **return** μ, B, scores $(x_t - \mu)^\top B$, weights W_t, σ

Proof. Define $H(\theta, \sigma) = \frac{1}{n}\sum_t g\big(r_t(\theta)^2/\sigma^2\big) - \delta$, so that $H(\theta, \hat{\sigma}(\theta)) = 0$. r_t^2 is a polynomial in (μ, B), and g is C^1 (Lemma 4.6), so H is C^1. Also $\partial_\sigma H = -\frac{2}{n\sigma^3}\sum_t g'(r_t^2/\sigma^2)\, r_t^2 < 0$ at θ^*, because some $r_t \in (0, \sigma^*)$ has $g' > 0$. By the implicit function theorem $\hat{\sigma}$ is C^1 near θ^*, with

$$\nabla_\theta \hat{\sigma} = -\frac{\nabla_\theta H}{\partial_\sigma H} = \frac{\sigma^*}{2\sum_t W_t^* r_t^2} \sum_t W_t^*\, \nabla_\theta r_t^2 = \frac{\sigma^*}{2\sum_t W_t^* r_t^2}\, \nabla_\theta L_{W^*}(\theta^*),$$

using $g'(\tau) = g'(0) w(\tau)$. Hence θ^* is stationary for $\hat{\sigma}$ iff it is stationary for L_{W^*} with frozen weights. For L_W: $\nabla_\mu r_t^2 = -2P_\perp(x_t - \mu)$ gives (5). For fixed μ, $L_W = \sum_t W_t \|\tilde{x}_t\|^2 - \text{tr}(B^\top C B)$; the Euclidean gradient of $\text{tr}(B^\top C B)$ is $2CB$, and its projection on the tangent space of $\mathcal{V}_{p,q}$ at B vanishes iff $P_\perp C B = 0$ (this is the standard first-order condition for trace maximisation on the Stiefel manifold), which is (6) after right-multiplying by $B^\top$. □

Remark 5.4. (6) holds for *every* q-dimensional invariant subspace of C^*. Among these, Corollary 2.5 says the top-q eigenspace is the global minimiser of L_{W^*}, which is the choice the algorithm makes.

5.3 Algorithm and monotone descent

Algorithm 1 restates the code of `pcaRobS` (`R/RobPCA_SM.R`, `RobStatTM` 1.0.13), which we read line by line. Lines 5–6 compute a global minimiser of L_W for the frozen weights (Corollary 2.5).

Theorem 5.5 (Monotone descent)**.** *Let g be nondecreasing and concave (Lemma 4.6). Fix θ_k with $\sigma_k = \hat{\sigma}(\theta_k)$ and weights $W_t = w(r_t(\theta_k)^2/\sigma_k^2)$. If θ_{k+1} satisfies*

$$L_W(\theta_{k+1}) \le L_W(\theta_k), \tag{7}$$

then $\hat{\sigma}(\theta_{k+1}) \le \sigma_k$.

Proof. Write $r_t = r_t(\theta_k)$ and $r_t' = r_t(\theta_{k+1})$. Concavity gives $g(b) \le g(a) + g'(a)(b - a)$, hence

$$\frac{1}{n}\sum_t g\Big(\frac{r_t'^2}{\sigma_k^2}\Big) \le \frac{1}{n}\sum_t g\Big(\frac{r_t^2}{\sigma_k^2}\Big) + \frac{g'(0)}{n\sigma_k^2}\sum_t W_t\,(r_t'^2 - r_t^2) = \delta + \frac{g'(0)}{n\sigma_k^2}\big\{L_W(\theta_{k+1}) - L_W(\theta_k)\big\} \le \delta.$$

So $h_{r'}(\sigma_k) \le \delta$, and Lemma 4.4 gives $\hat{\sigma}(r') \le \sigma_k$. □

Corollary 5.6. *In Algorithm 1 the sequence σ_k is nonincreasing and converges. Any limit point θ^* of the iterates at which the conditions of Theorem 5.3 hold and that is a fixed point of the update is a stationary point of $\hat{\sigma}$.*

Proof. Lines 5–6 minimise L_W globally, so (7) holds and Theorem 5.5 applies; a nonincreasing sequence bounded below by 0 converges. At a fixed point, (μ^*, B^*) minimises L_{W^*}, so its stationarity conditions hold, which by Theorem 5.3 are those of $\hat{\sigma}$. □

Remark 5.7 (What is not guaranteed). $\hat{\sigma}(\theta)$ is not convex; the iteration can stop at a local minimiser. This is why the start must already be robust. The spherical start gives each day unit norm, so one large day carries weight $1/n$. Convergence of the iterates themselves, as opposed to the scale values, is not claimed.

5.4 Explained variance and component variances

Let $\sigma_0 = \hat{\sigma}(\|x_1 - \mu^0\|, \ldots, \|x_n - \mu^0\|)$ be the M-scale of the centred norms. `pcaRobS` reports the robust proportion of explained variability

$$\widehat{\mathrm{PEV}} = 1 - \hat{\sigma}(\hat{\theta})^2/\sigma_0^2.$$

Robust variances of the individual components are best taken from the robust scatter of the scores (Section 6.2). The eigenvalues of $C_W / \sum_t W_t$ are biased downwards, because the weights shrink clean days in the tails of the distribution more than central ones.

6 Limits of residual-based weights

6.1 The individual components are not protected

Proposition 6.1. *Let the weights be any function of the residual, $W_t = \phi(r_t)$, with $\phi(0) = \max \phi$. If $x_{t_o} - \hat{\mu} \in \mathrm{span}(\hat{B})$ then $r_{t_o} = 0$ and day t_o receives the maximal weight, whatever its size. Its contribution to C_W is $\phi(0)\, \tilde{x}_{t_o} \tilde{x}_{t_o}^\top$, with norm $\phi(0)\|\tilde{x}_{t_o}\|^2$, which is unbounded.*

Combining this with Lemma 3.3: an in-span day does not tilt the S-PCA subspace, but it rotates the eigenvectors of C_W inside it. Because clean days have weights strictly below 1 (mean 0.62 in our simulation) while the in-span day has weight 0.92, its *relative* weight is *larger* than under classical PCA. Its share of $\mathrm{tr}\, C_W$ was 0.29, against 0.21 for $\mathrm{tr}\, S$. Table 2 confirms this: for an in-span shock, S-PCA gives PC1/PC2 errors of about 18°, against 13° for classical PCA, while its subspace error is unchanged.

6.2 Second stage: robust rotation inside the subspace

Let $(\hat{\mu}, \hat{B})$ be the S-PCA estimate and $s_t = \hat{B}^\top (x_t - \hat{\mu}) \in \mathbb{R}^q$ the scores. Let $\hat{\Sigma}_s$ be an affine-equivariant robust scatter estimator in $\mathbb{R}^q$. We use the reweighted MCD (Rousseeuw and Van Driessen, 1999); with $q = 3$ and n in the hundreds this is cheap and has high breakdown. With $\hat{\Sigma}_s = \hat{V} \hat{\Lambda} \hat{V}^\top$, define the final loadings and robust component variances as

$$\hat{A} = \hat{B}\hat{V}, \qquad \hat{\lambda}_k^{\mathrm{rob}} = \hat{\Lambda}_{kk}. \tag{8}$$

Proposition 6.2 (Well-posedness)**.** *$\hat{A}$ does not depend on the choice of orthonormal basis of* $\mathrm{span}(\hat{B})$.

Proof. Replacing $\hat{B}$ by $\hat{B}R$ with $R \in O(q)$ maps scores to $R^\top s_t$. Affine equivariance gives $\hat{\Sigma}_s \mapsto R^\top \hat{\Sigma}_s R$, whose eigenvectors are $R^\top \hat{V}$. Hence $\hat{A} \mapsto \hat{B} R R^\top \hat{V} = \hat{A}$. □

The in-span day has a large Mahalanobis distance in score space, so the MCD downweights it there. This is the step that residual weights cannot perform.

7 Huber-type and norm-capping weights: w or w^2?

A common heuristic caps each day's norm with $w_t = \min(1, c/\|\tilde{x}_t\|)$. This section makes precise which power of this weight belongs in the scatter matrix, and why.

Every IRLS scheme above solves, at its fixed point, $P_\perp C_W P = 0$ (Theorem 5.3), in which day t enters through

$$T_t = W_t\, P_\perp \tilde{x}_t \tilde{x}_t^\top P, \qquad \|T_t\| = W_t\, r_t\, \|P\tilde{x}_t\|,$$

and the within-subspace eigenvectors are driven by $W_t\, P\tilde{x}_t\tilde{x}_t^\top P$, of norm $W_t\|P\tilde{x}_t\|^2$. Take an outlier $\tilde{x} = s\,u$ with $\|u\| = 1$ and let $s \to \infty$ at a fixed (μ, B).

Proposition 7.1. *Assume $Pu \neq 0$ and $P_\perp u \neq 0$ (an off-span outlier that also loads on the components).*

(a) Classical ($W \equiv 1$): $\|T\| = s^2\|P_\perp u\|\|Pu\|$, quadratic in s.

(b) Huber M-estimator of the residuals with fixed scale σ: $\rho_H(u) = u^2/2$ for $|u| \le c$ and $c|u| - c^2/2$ otherwise. Here g_H is concave, so Theorem 5.5 applies with fixed scale, and the IRLS weight is $W = \min(1, c\sigma/r)$. Then $\|T\| = \min(r, c\sigma)\|P\tilde{x}\| = c\sigma s\|Pu\|$ for large s, linear in s. This is the leverage effect familiar from Huber regression: ψ is bounded but the leverage $\|P\tilde{x}\|$ is not.

(c) Bisquare residual weights (S-PCA): $T = 0$ once $r = s\|P_\perp u\| \ge \sigma$.

(d) Norm weights $W = \min(1, c/\|\tilde{x}\|)^m$: for large s, $\|T\| = c^m s^{2-m}\|P_\perp u\|\|Pu\|$. This is bounded (by $c^2/2$) for $m = 2$ and linear in s for $m = 1$. The within-subspace term $W\|P\tilde{x}\|^2$ behaves the same way.

Proof. Direct substitution. Use $r = s\|P_\perp u\|$, $\|P\tilde{x}\| = s\|Pu\|$ and $r\|P\tilde{x}\| \le \frac{1}{2}(r^2 + \|P\tilde{x}\|^2) = \frac{1}{2}\|\tilde{x}\|^2$. For (b), $g_H(\tau) = \tau/2$ on $[0, c^2]$ and $c\sqrt{\tau} - c^2/2$ beyond, so $g_H'(\tau) = \frac{1}{2}\min(1, c/\sqrt{\tau})$ is nonincreasing. □

Remark 7.2 (Answer to "w or w^2"). With norm-based capping $w_t = \min(1, c/\|\tilde{x}_t\|)$, the scatter matrix must be built as $\sum_t w_t^2 \tilde{x}_t\tilde{x}_t^\top$, i.e. rows scaled by w_t. In Bailey's notation the weight is $W_{tj} = w_t^2$. This is a Huber-type M-estimator of scatter whose weight $u(d^2) = \min(1, c^2/d^2)$ is a function of the *squared* distance with $d^2u(d^2)$ bounded, as in Maronna (1976), but with the shape matrix fixed at I. Using $W_{tj} = w_t$ leaves a contribution that grows linearly in $\|\tilde{x}_t\|$ (Table 2: 5.8° against 2.1°). Norm weights also downweight in-span days, but they never redescend to zero, so they are not a high-breakdown method. The spherical limit $u(d^2) = 1/d^2$ gives every day the same unit contribution (Locantore et al., 1999).

8 Implementation by weighted EMPCA

8.1 Bailey's weighted EMPCA, as implemented

Bailey (2012) fits, to data that have been *centred beforehand*, the model $x_i \approx \sum_k c_{ik}\phi_k$ by minimising

$$\chi^2(C, \Phi) = \sum_{i=1}^{n}\sum_{j=1}^{p} \omega_{ij}\Big(x_{ij} - \sum_{k=1}^{q} c_{ik}\phi_{kj}\Big)^2, \tag{9}$$

where $\omega_{ij} \ge 0$ are inverse noise variances and $\omega_{ij} = 0$ encodes a missing value. Here $\Phi \in \mathbb{R}^{q\times p}$ has rows ϕ_k (so $B = \Phi^\top$) and $C \in \mathbb{R}^{n\times q}$. From the reference code (`empca.py`, `https://github.com/sbailey/empca`), one iteration is:

(1) *E-step* (`solve_coeffs`): for each i, $c_i = (\Phi\Omega_i\Phi^\top)^{-1}\Phi\Omega_i x_i$, with $\Omega_i = \mathrm{diag}(\omega_{i\cdot})$. If $\omega_{i\cdot}$ is constant across j, the code uses $c_i = \Phi x_i$, which is equivalent because Φ is orthonormal.

(2) *M-step* (`solve_eigenvectors`): for $k = 1, \ldots, q$ in turn, $\phi_{kj} \leftarrow \sum_i \omega_{ij} c_{ik} x_{ij}^{(k)} / \sum_i \omega_{ij} c_{ik}^2$ for all j. Then deflate, $x_i^{(k+1)} = x_i^{(k)} - c_{ik}\phi_k$ (starting from $x^{(1)} = x$).

(3) Gram–Schmidt orthonormalisation of $\phi_1, \ldots, \phi_q$.

The code starts from random orthonormal vectors and runs a fixed number of iterations (25 by default); it does not estimate a mean. A joint alternative (`lower_rank` in the same file, the criss-cross regression of Gabriel and Zamir, 1979) solves, for each j, the weighted least-squares problem for the whole column $\phi_{\cdot j}$ at once.

8.2 Row-constant weights reproduce the weighted eigenproblem

Proposition 8.1. *Let $\omega_{ij} = W_i \geq 0$ (one weight per day), let the data be centred, let $C_W = \sum_i W_i x_i x_i^\top$, and suppose all denominators in step (2) are nonzero. Then:*

(a) the global minimisers of (9) are those with $\mathrm{span}(\Phi^\top)$ equal to the top-q eigenspace of C_W (if $\lambda_q > \lambda_{q+1}$);

(b) an orthonormal q-tuple $(\phi_1, \ldots, \phi_q)$ is a fixed point of Bailey's iteration (1)–(3) iff each ϕ_k is an eigenvector of C_W with a positive eigenvalue.

Proof. (a) With $\omega_{ij} = W_i$, $\chi^2 = \sum_i W_i \|x_i - \Phi^\top c_i\|^2$. For fixed Φ the optimal c_i is the projection, so $\min_C \chi^2 = L_W(0, \Phi^\top)$, and Corollary 2.5 with $\mu = 0$ (the data are centred) applies. (b) The E-step gives $c_{ik} = \phi_k^\top x_i$, so the step-(2) numerator for $k = 1$ is $\sum_i W_i c_{i1} x_i = C_W\phi_1$ and the denominator is $\phi_1^\top C_W \phi_1$. *If* $C_W\phi_k = \lambda_k\phi_k$ for all k, then ϕ_1 is unchanged. Inductively, $x^{(k)} = (I - \sum_{l<k}\phi_l\phi_l^\top)x$, so the update is $(I - \sum_{l<k}\phi_l\phi_l^\top)C_W\phi_k/\lambda_k = \phi_k$; Gram–Schmidt leaves an orthonormal set unchanged. *Conversely*, at a fixed point the normalised update of ϕ_1 is $C_W\phi_1/\|C_W\phi_1\| = \phi_1$, so ϕ_1 is an eigenvector. If $\phi_1, \ldots, \phi_{k-1}$ are eigenvectors and $\phi_k \perp \phi_l$ for $l < k$, then $(I - \sum_{l<k}\phi_l\phi_l^\top)C_W\phi_k = C_W\phi_k$, because $\phi_l^\top C_W \phi_k = \lambda_l \phi_l^\top \phi_k = 0$. The fixed-point condition then makes ϕ_k an eigenvector. □

So with one weight per day, Bailey's EMPCA computes the eigenvectors of the weighted scatter. This is lines 5–6 of Algorithm 1, provided the centring is done with the same weights. The iteration is a deflated power method; its fixed points include all eigenvector tuples, and in practice it converges to the leading ones, which we verify numerically (0.004° from Algorithm 1, Table 1).

8.3 Robust EMPCA

General ω_{tj}, for heteroskedastic tenors or missing quotes, require a residual norm that is comparable across days with different numbers $m_t = \#\{j : \omega_{tj} > 0\}$ of observed tenors. We define

$$\bar\omega_{tj} = \frac{p}{m_t}\,\omega_{tj}, \qquad r_t(\mu, \Phi, c_t)^2 = \sum_{j=1}^{p} \bar\omega_{tj}\Big(x_{tj} - \mu_j - \sum_k c_{tk}\phi_{kj}\Big)^2. \tag{10}$$

If the residuals are homoscedastic noise with variances $1/\omega_{tj}$, then $\mathbb{E}r_t^2 = p$ for every t, whatever m_t. When $\omega \equiv 1$ and c_t is the projection, (10) reduces to (1). The S-objective is $\hat\sigma(r_1, \ldots, r_n)$ minimised over (μ, Φ, C), and the frozen-weight criterion is

$$L_W(\mu, \Phi, C) = \sum_t \sum_j W_t\,\bar\omega_{tj}\Big(x_{tj} - \mu_j - \sum_k c_{tk}\phi_{kj}\Big)^2, \tag{11}$$

which is Bailey's χ^2 with weights $W_t\bar\omega_{tj}$, plus a centre.

Algorithm 2 Robust EMPCA (outer M-scale reweighting, inner weighted EMPCA)

Require: X, ω, q, δ, inner sweeps m (we use 5), tolerance ϵ

$\mu_j^0 =$ median of observed $x_{\cdot j}$; $z_t = (x_t - \mu^0)/r_t^0$ on observed cells, with r_t^0 from (10) at $\Phi = 0$

$\Phi \leftarrow$ Bailey EMPCA of z with weights ω ▷ spherical start

E-step on $x - \mu^0$; $r_t \leftarrow$ (10); $\sigma \leftarrow \hat{\sigma}(r)$

repeat

  $W_t \leftarrow w(r_t^2/\sigma^2)$ ▷ outer step: frozen day weights

  for m sweeps **do** ▷ inner step: each update lowers L_W

    centre: $\mu_j \leftarrow \sum_t W_t \bar{\omega}_{tj}(x_{tj} - (\Phi^\top c_t)_j) / \sum_t W_t \bar{\omega}_{tj}$

    E-step: $c_t \leftarrow (\Phi \bar{\Omega}_t \Phi^\top)^{-1} \Phi \bar{\Omega}_t (x_t - \mu)$ ▷ W_t cancels

    M-step: *exact* $\phi_{\cdot j} \leftarrow (C^\top D_j C)^{-1} C^\top D_j (x_{\cdot j} - \mu_j)$, $D_j = \mathrm{diag}(W_t \bar{\omega}_{tj})$; then QR-orthonormalise $\Phi^\top$

      (or Bailey's sequential step (2)–(3), see Remark 8.3)

    E-step again

  end for

  $r_t \leftarrow$ (10); $\sigma_{\mathrm{new}} \leftarrow \hat{\sigma}(r)$

until $|\sigma - \sigma_{\mathrm{new}}| \le \epsilon\sigma$

Stage 2 (Section 6.2): robust scatter of the scores c_t, rotate $\Phi^\top$ to $\hat{A}$

Theorem 8.2 (Descent for robust EMPCA)**.** *In Algorithm 2 with the exact M-step, the outer scales satisfy* $\sigma_{k+1} \le \sigma_k$.

Proof. With W frozen, each inner update minimises L_W exactly over one block. The centre update minimises over μ (per column, a weighted mean). The E-step minimises over each c_t (a weighted least-squares problem; W_t is a common factor of the row and cancels). The exact M-step minimises over each column $\phi_{\cdot j}$. The QR step replaces $\Phi^\top$ by $Q = \Phi^\top R^{-1}$ and, through the following E-step, leaves $\min_C L_W$ unchanged because it preserves $\mathrm{span}(\Phi^\top)$. So L_W is nonincreasing across the inner loop, which is (7) in the variables (μ, Φ, C). The proof of Theorem 5.5 goes through verbatim with r_t from (10). □

Remark 8.3 (Bailey's sequential M-step). Step (2) updates ϕ_k against data deflated by components $l < k$ only; it is not a block minimisation of (9). So Theorem 8.2 does not cover it. With missing values we observed small increases of L_W (up to 6×10^{-4} relative) and of σ (up to 3×10^{-4}) in 8 of 10 runs, with no effect on accuracy (Table 3). With one weight per day and no missing data, Proposition 8.1 shows Bailey's step targets the right eigenvectors. To keep the guarantee in general, use the exact column update, or accept a Bailey step only if L_W decreases.

Remark 8.4 (What changes relative to plain Bailey EMPCA). (i) Bailey's weights become $W_t \bar{\omega}_{tj}$, with W_t recomputed in an outer loop from the M-scale of the residual norms. (ii) The centre is re-estimated inside the loop with the same weights; it is *not* the raw column mean, which the large day would contaminate. (iii) The start is spherical, not random, so that the first weights are already robust. (iv) With general ω, (11) is non-convex and may have several local minima (Srebro and Jaakkola, 2003), which is another reason for a robust start. (v) The final basis is rotated by Stage 2; with the exact M-step the QR basis is arbitrary inside the subspace anyway. (vi) Bailey's R^2 is not robust (the code's docstring says so), so report $\widehat{\mathrm{PEV}}$ or the Stage-2 variances.

Remark 8.5 (Related estimators). Maronna and Yohai (2008) fit a robust rank-q approximation by alternating M-regressions with *cell-wise* weights. That is the natural extension if individual quotes, rather than whole days, are bad. Boente and Salibián-Barrera (2015) minimise a sum of coordinate-wise M-scales of the residuals. Projection-pursuit (Croux and Ruiz-Gazen, 2005) and ROBPCA (Hubert et al., 2005) are useful benchmarks.

9 Numerical verification

Design. $p = 10$ tenors $\tau \in \{0.25, 0.5, 1, 2, 3, 5, 7, 10, 20, 30\}$; true loadings L are the orthonormalised Nelson–Siegel level, slope and curvature vectors; $x_t = Lf_t + e_t$ with $f_t \sim N(0, \mathrm{diag}(6^2, 3^2, 1.5^2))$, $e_t \sim N(0, 0.4^2 I)$, $n = 750$, $q = 3$. Day 400 is replaced by 18× the median centred norm times a unit vector u. For an *off-span* shock, $u \propto (1, 1, 1, 1, -0.3, \ldots, -1)$; for an *in-span* shock, $u \propto L_{\cdot 1} + L_{\cdot 2}$ (level plus slope). Errors are the largest principal angle between span($\hat{B}$) and span(L) ("subspace") and the angles between matched individual loadings ("PC1–PC3"), in degrees, over 30 replications. Python code, the `RobStatTM` 1.0.13 source, and Bailey's `empca.py` were run in the same environment.

Table 1: Validation (one data set, off-span shock).

Comparison	Angle (deg.)
Algorithm 1 (Python) vs. `RobStatTM::pcaRobS` (R), subspace	7×10^{-6}
Algorithm 2 (Bailey inner step) vs. Algorithm 1, subspace	0.004
Weight given to the shock day (both algorithms)	0
σ path of Algorithm 1: 2.2445 → 2.2395 → 2.2394 → 2.2394	monotone

Table 2: Monte Carlo errors (degrees): mean over 30 replications, with the maximum in parentheses for the subspace.

	Off-span shock				In-span shock (level+slope)			
Method	Subspace	PC1	PC2	PC3	Subspace	PC1	PC2	PC3
Classical PCA	13.06 (14.4)	2.42	15.27	15.76	1.57 (2.1)	13.29	13.31	1.77
Classical, day removed	1.60 (2.1)	1.35	1.76	1.94	1.60 (2.1)	1.35	1.76	1.94
Norm cap, $W = w$	5.79 (7.0)	1.47	6.54	6.98	1.59 (2.1)	3.40	3.56	1.86
Norm cap, $W = w^2$	2.06 (2.9)	1.37	2.36	2.55	1.59 (2.1)	1.56	1.93	1.92
S-PCA (Algorithm 1)	1.66 (2.3)	1.37	1.91	2.09	1.63 (2.3)	18.35	18.37	1.86
S-PCA + Stage 2	1.66 (2.3)	1.76	2.52	2.36	1.63 (2.3)	1.79	2.43	2.37
Robust EMPCA (Algorithm 2)	1.66 (2.3)	1.38	1.92	2.09	1.63 (2.3)	18.31	18.33	1.86
Robust EMPCA + Stage 2	1.66 (2.3)	1.76	2.52	2.36	1.63 (2.3)	1.79	2.43	2.37

Table 3: 10% of cells missing at random (off-span shock, 10 replications); subspace error in degrees.

Method	Mean	Max	σ path
Bailey EMPCA, non-robust	13.40	–	–
Robust EMPCA, Bailey M-step	1.61	2.23	up slightly, 8/10 runs
Robust EMPCA, exact M-step	1.59	2.22	monotone, 10/10

Reading the tables. (1) The robust EMPCA of Algorithm 2 matches `pcaRobS` and gives the shock day weight 0. (2) Off-span shocks are fully handled by residual-based methods. In-span shocks are not, at the level of individual components: Stage 2 is needed and brings the errors back to within about 1° of an oracle that deletes the day. With the MCD it costs a little accuracy on off-span data (about 0.4° per component); a more efficient robust scatter (e.g. an MM-estimator) would reduce that. (3) Norm capping with $W = w^2$ is a good, simple estimator in both cases; $W = w$ removes only about half of the distortion. (4) The robustness of $\delta = 0.5$ costs little efficiency here; with one known outlier, $\delta = 0.25$ is equally safe and slightly more efficient.

Table 4: Efficiency on clean data (no shock), mean subspace error over 30 replications.

	Classical	Norm cap w^2	S-PCA $\delta = 0.1$	$\delta = 0.25$	$\delta = 0.5$
Error (deg.)	1.599	1.599	1.599	1.605	1.661

10 Recommended procedure

1. Check the large day for data errors (roll, rebase, stale print). Robust estimation is not a substitute for clean data.
2. Estimate the subspace by Algorithm 2. Use Bailey weights $W_t\bar{\omega}_{tj}$, an exact (or safeguarded) M-step, a weighted centre inside the loop, a spherical start, and $\delta \in [0.25, 0.5]$.
3. Rotate inside the subspace with a robust scatter of the scores (Stage 2, Section 6.2). Report $\hat{A}$ and $\hat{\lambda}^{\mathrm{rob}}$.
4. Report $\widehat{\mathrm{PEV}}$, the day weights W_t (the large day should be near 0 if off-span), and, for comparison with legacy numbers, the classical variances of the raw data projected on $\hat{A}$.
5. Cross-check the subspace against `RobStatTM::pcaRobS` on the complete-data days.

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